



HUMAN-COMPUTER INTERACTION

EcoGo

Travel Green · Live Clean

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1 Problem/Solution Overview

We identified two main issues in sustainable tourism:

1. Users want to be sustainable, but eco-friendly options are often inconvenient or hard to find.
2. Travel decisions are usually driven by cost and convenience rather than environmental impact.

To solve this, we created **EcoGo**, a smart travel companion. It addresses these problems through:

- **Ease of Planning:** Highlighting low-impact routes and bundles (e.g., comparing train vs. plane).
- **Gamification:** Using points, leaderboards, and AI tips to make sustainable travel fun and rewarding.

2 Needfinding

2.1 Domain

We focused on **environmentally sustainable tourism**, specifically promoting habits like walking, using public transport, and supporting local communities to reduce pollution.

2.2 Interviews

We interviewed 4 participants to understand their habits:

- **Participant 1 Carafassi Samuele (UniPd Cybersecurity) (Immediate user):** Cost-driven but motivated by gamification and badges.
- **Participant 2 Farinelli Simone (University of Bristol student)(Immediate user):** Values ecosystems but prioritizes simplicity and safety in apps like Google Maps.
- **Participant 3 Isa Yaqoubi (Biomedical engineering students at the University of Bologna) (Extreme User):** Strictly budget-driven; walks everywhere, which naturally aligns with sustainability.
- **Participant 4 Ramzy cherif (working at the National Erasmus+ Office) (Immediate user):** Believes sustainable tourism is a "win for everyone" and focuses on organized planning.

2.3 Synthesis

From the interviews, we identified five deep user needs:

1. **Simplicity:** Eco-friendly choices must be easy to find.
2. **Control:** Users need clear info to feel safe and confident.
3. **Transparency:** Users want to trust how their data is used.
4. **Authenticity:** Users want real local experiences, not tourist traps.
5. **Motivation:** Users enjoy friendly, non-stressful competition.

3 Solutions

3.1 Chosen Solution

We selected a **Web Application**. This format allows users to easily plan trips, visualize their carbon footprint, and access the platform from any device without specialized hardware.

We have also considered a **Smart glasses solution**, but we don't have experience and materials.

3.2 Value Proposition

"Travel Green · Live Clean"

The platform integrates planning and exploration into a seamless experience, making sustainable behavior easy and effortless.

4 Tasks and Storyboard

4.1 Tasks

We designed the system around three core tasks:

1. **Simple Task (Discover):** Quickly finding sustainable travel options.
2. **Moderate Task (Plan):** Organizing a trip that balances budget, fun, and sustainability.
3. **Complex Task (Track):** Monitoring impact (CO_2 saved) and sharing experiences.

4.2 Storyboard

Our storyboard follows a user who plans a trip, selects eco-friendly tools, shares their experience, and receives a notification celebrating the emissions they saved.



Figure 1: Scenario of web app solution

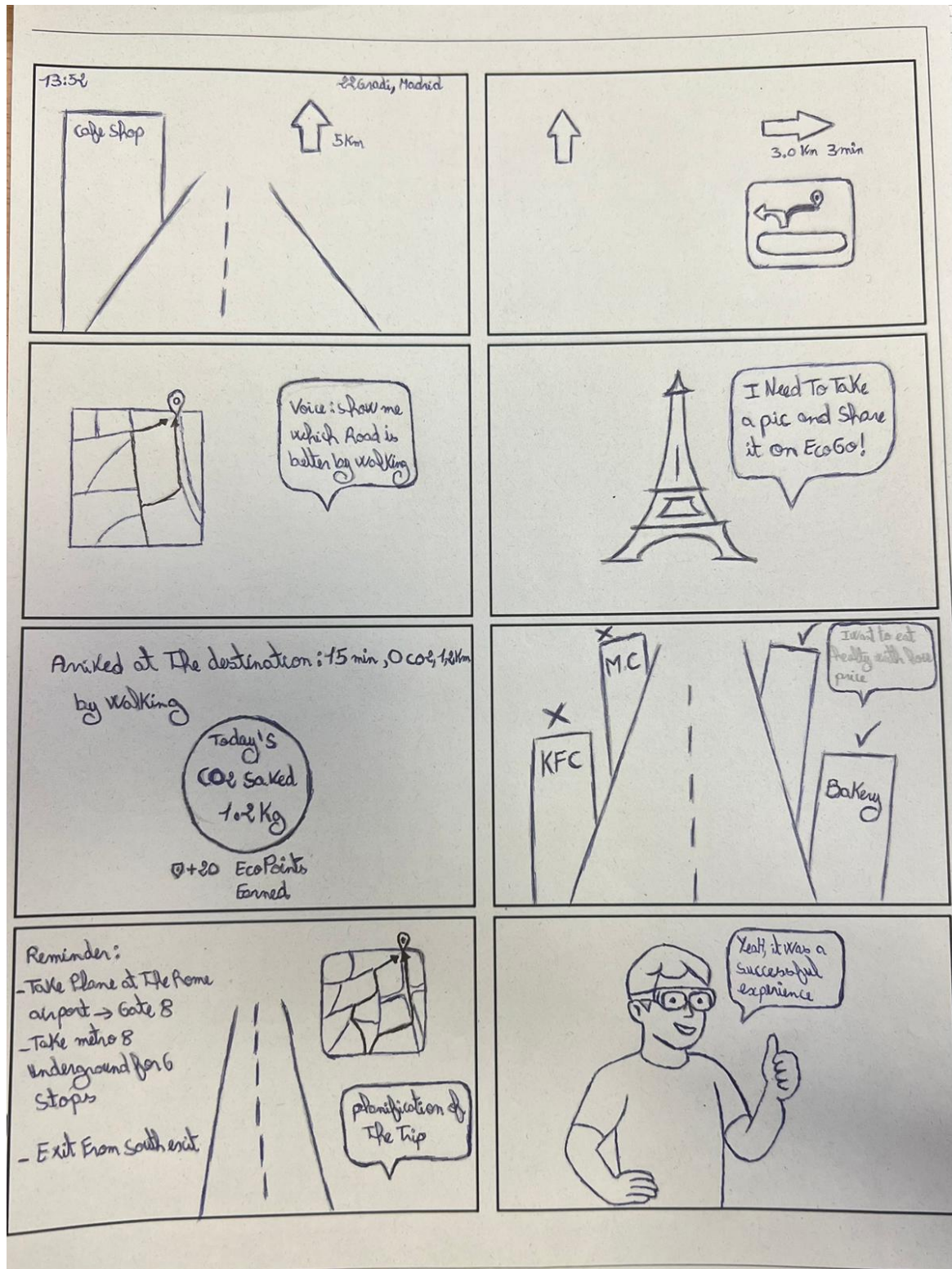


Figure 2: Scenario of smart glasses solution

5 Low-fidelity Prototypes

5.1 Modality Exploration

We considered two approaches:

- **Web Application (Chosen):** Best for centralizing travel costs and detailed sustain-

ability info.

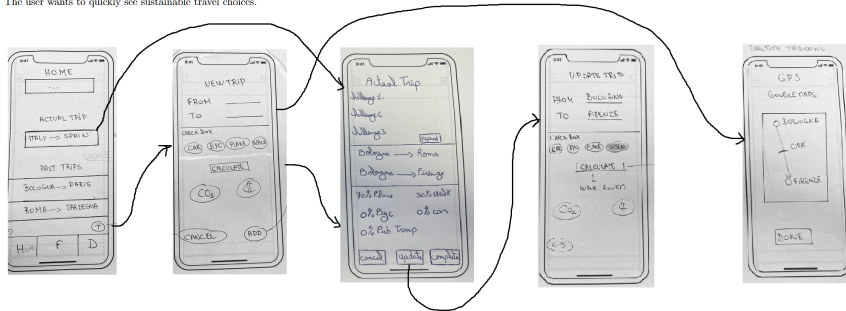
- **Smart Glasses (Discarded):** An AR experience that was too complex and required expensive hardware.

5.2 Paper Prototypes

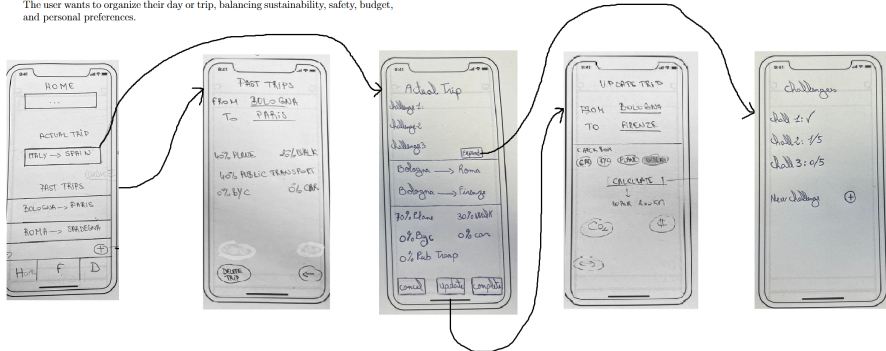
We sketched the Web App flow to support our main tasks:

- **Home/New Trip:** Entering destination details.
- **Eco-Score Dashboard:** Viewing leaderboards and saved CO_2 .
- **Trip View:** visualizing the route and leaving reviews.

- **1. Simple Task: Discover Eco-Friendly Options**
The user wants to quickly see sustainable travel choices.



- **2. Moderate Task: Plan a Safe and Enjoyable Trip**
The user wants to organize their day or trip, balancing sustainability, safety, budget, and personal preferences.



- **3. Complex Task: Track, Reflect, and Share Sustainable Actions**
The user wants to monitor their sustainable choices, see their impact, complete optional challenges, and share experiences with others.

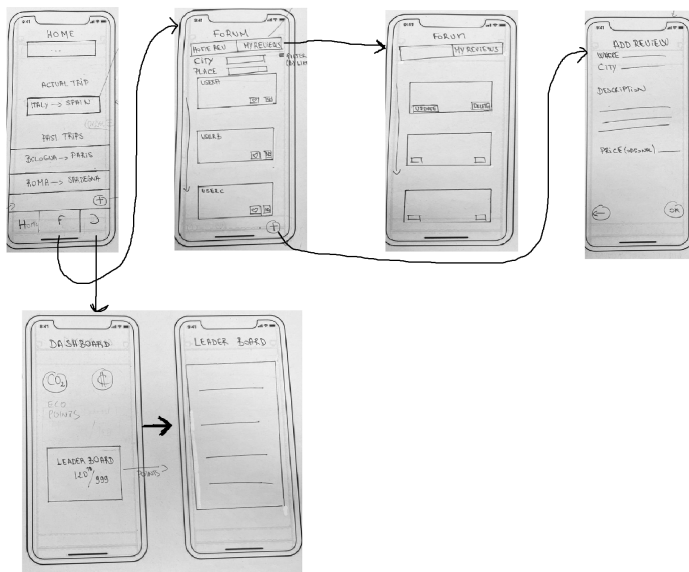


Figure 3: Web flow of web app solution

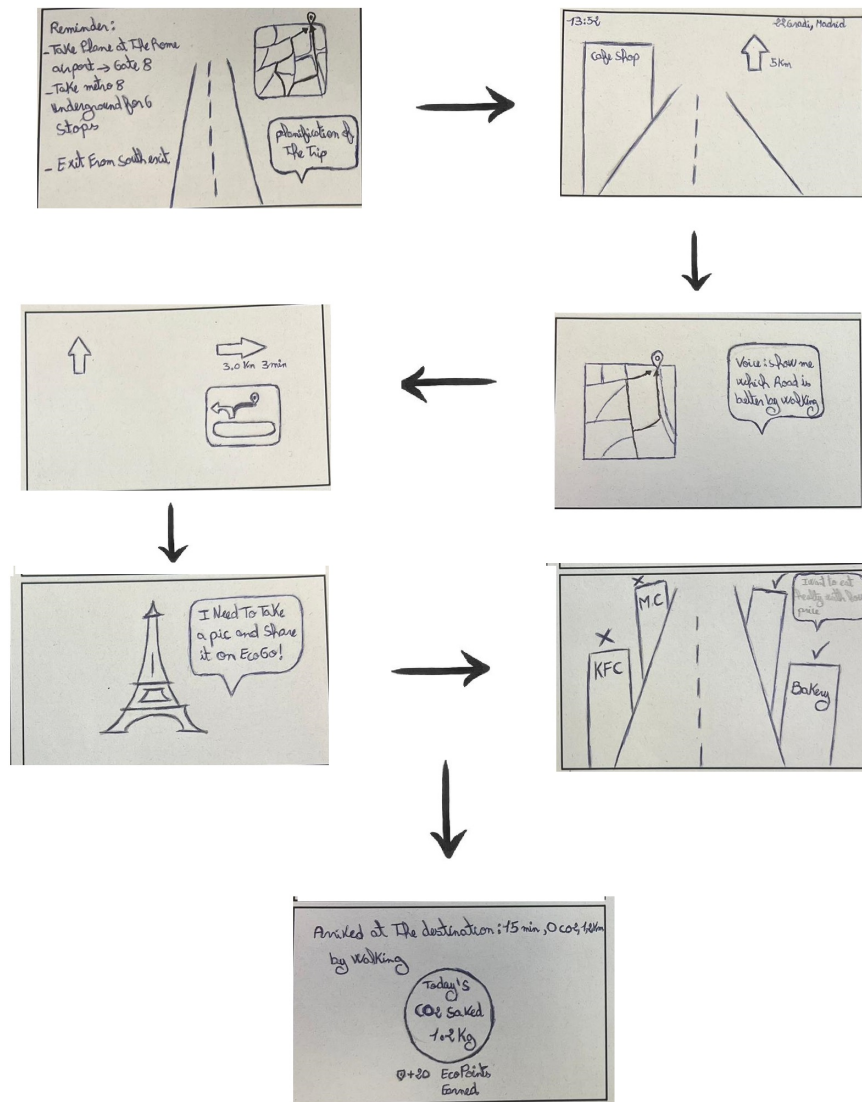


Figure 4: Web flow of smart glasses solution

6 Medium to High-Fidelity Prototype

6.1 Prototype Selection

We used **Figma** to refine two critical screens:

- **Dashboard:** Displays "Total CO₂ Saved," "Eco Score," and trend graphs to motivate the user.
- **Planning Page:** Allows users to input travel details to calculate their footprint.



EcoVoyage
Travel Green, Live Clean



Total CO₂ Saved

163 kg

↗ 15% vs last month

Green Miles

2,847

🌱 1,200 this month

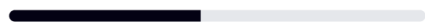
Eco Score

892

🏆 Top 10% globally

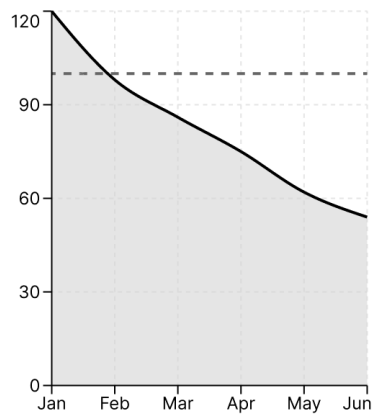
Monthly Goal

54 / 100 kg



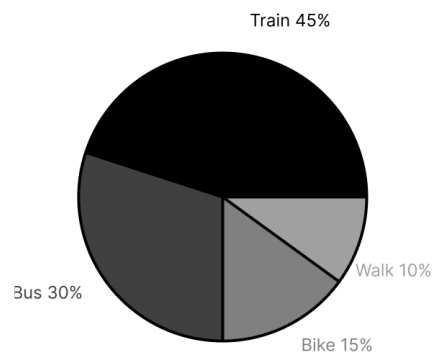
Carbon Footprint Trend

Monthly CO₂ emissions (kg)



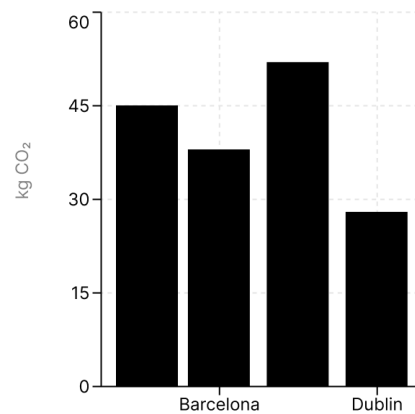
Transport Modes

Your eco-friendly choices



CO₂ Saved by Destination

Compared to conventional travel methods



🌍 Your Environmental Impact



Equivalent to planting

7 trees



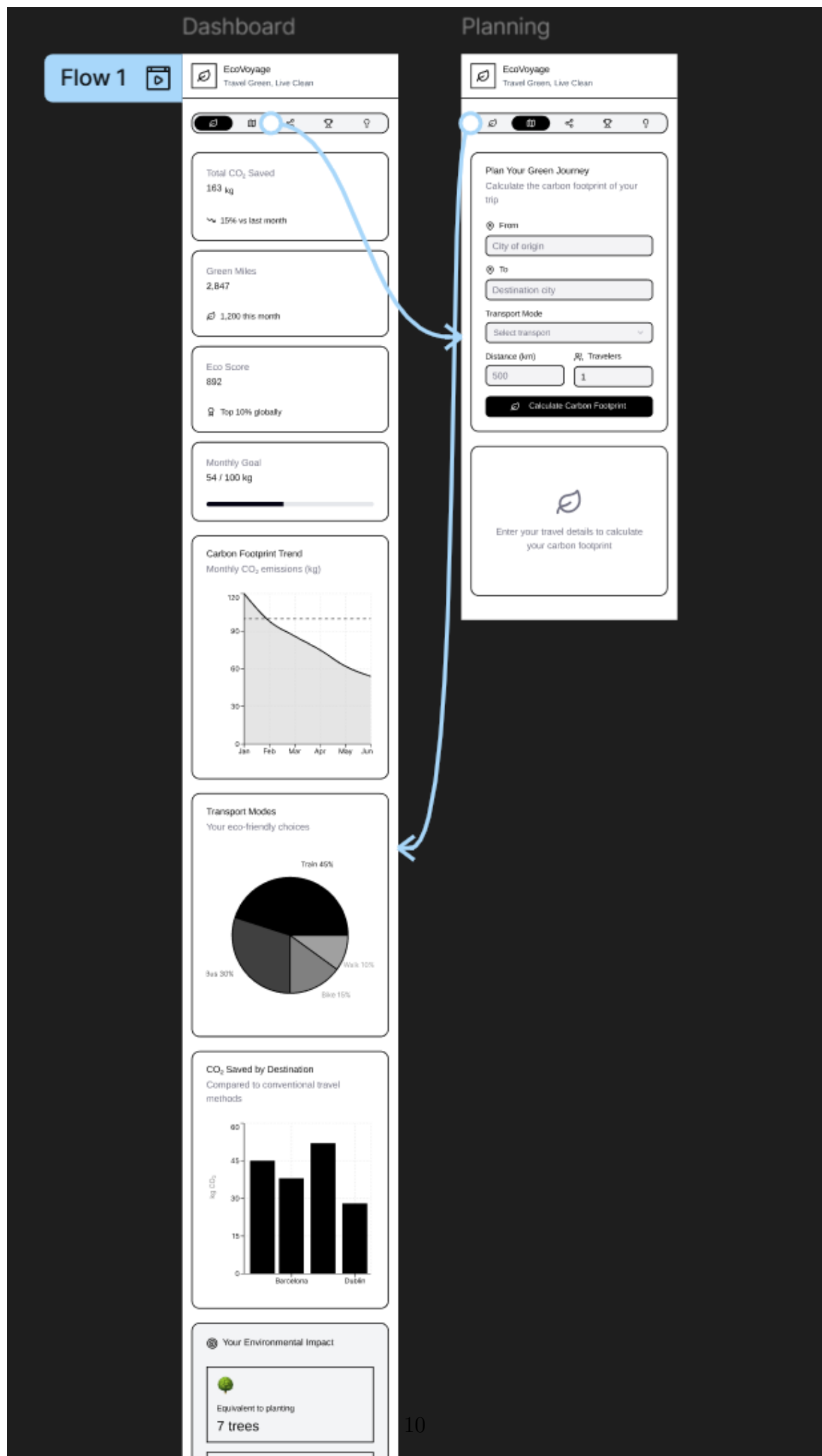
Energy saved equals

340 kWh



Car miles avoided

458 miles





EcoVoyage

Travel Green, Live Clean



Plan Your Green Journey

Calculate the carbon footprint of your trip

📍 From

📍 To

Transport Mode



Distance (km)

👤 Travelers



Calculate Carbon Footprint

7 High-fidelity Prototype

7.1 Technology Stack

We built a responsive Web Application using the **MEVN stack**:

- **Frontend:** Vue 3 + Vite with TailwindCSS, DaisyUI, Lucide.vue and Turf.js for styling.
- **Backend:** Node.js + Express with MongoDB for data storage.
- **APIs:** Mapbox for maps and Google Gemini AI for sustainable travel tips.

7.2 Main functionality

- **Tips:** Users receive daily tips generated by AI.
- **Plan:** Users compare emissions (e.g., Train vs. Car) on an interactive map.
- **Dashboard and Rewards:** The Dashboard displays "Eco-Points" to reward good choices. There is also a leaderboard for global ranking, used to motivate green actions.
- **Notification:** The live page notify weather or traffic alerts
- **Past trips memorization:** Maybe one day user wants to check the past memorable trips.
- **Feedback:** There is a feedback dedicated page in which users can make suggestions for better usability. The problems will send to admin page.
- **Languages switching and light/dark theme:** These features increase usability of the system.
- **Discover:** The discover page users can share and find interesting activities/places.

7.3 Visual Design

We used a nature-inspired palette (Emerald and Forest Green) and a card-based layout to make complex data easy to read.

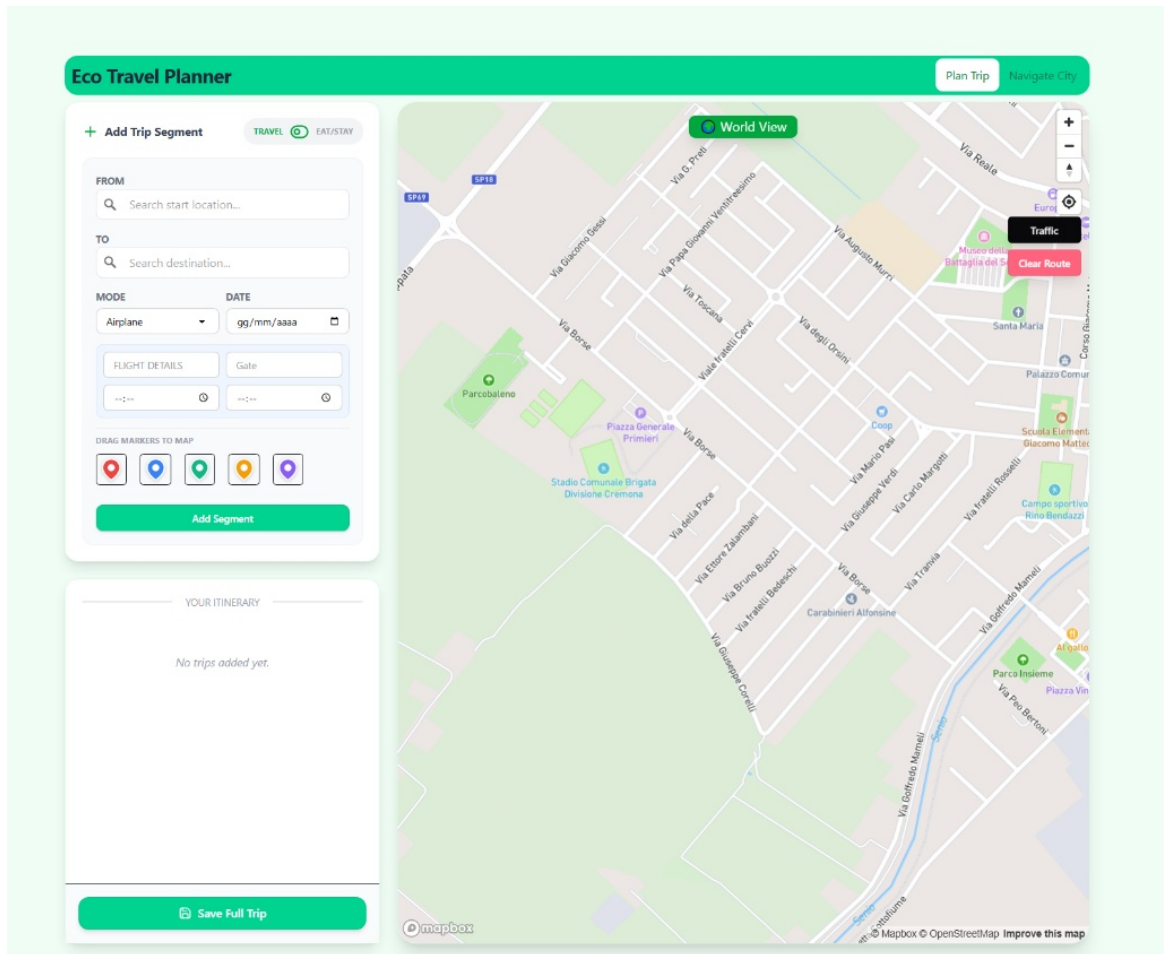


Figure 8: High-Fidelity Planning Map Interface

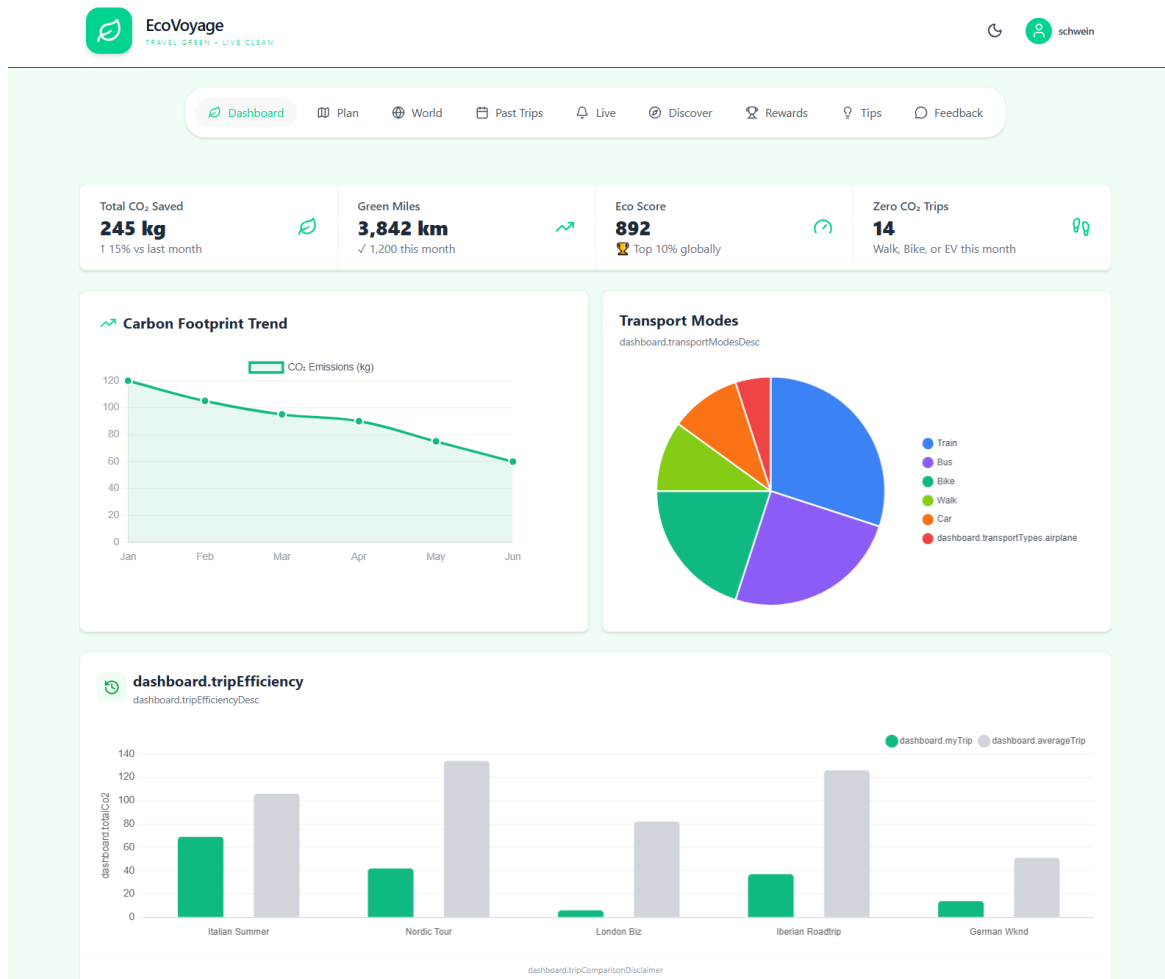


Figure 9: High-Fidelity Dashboard Interface

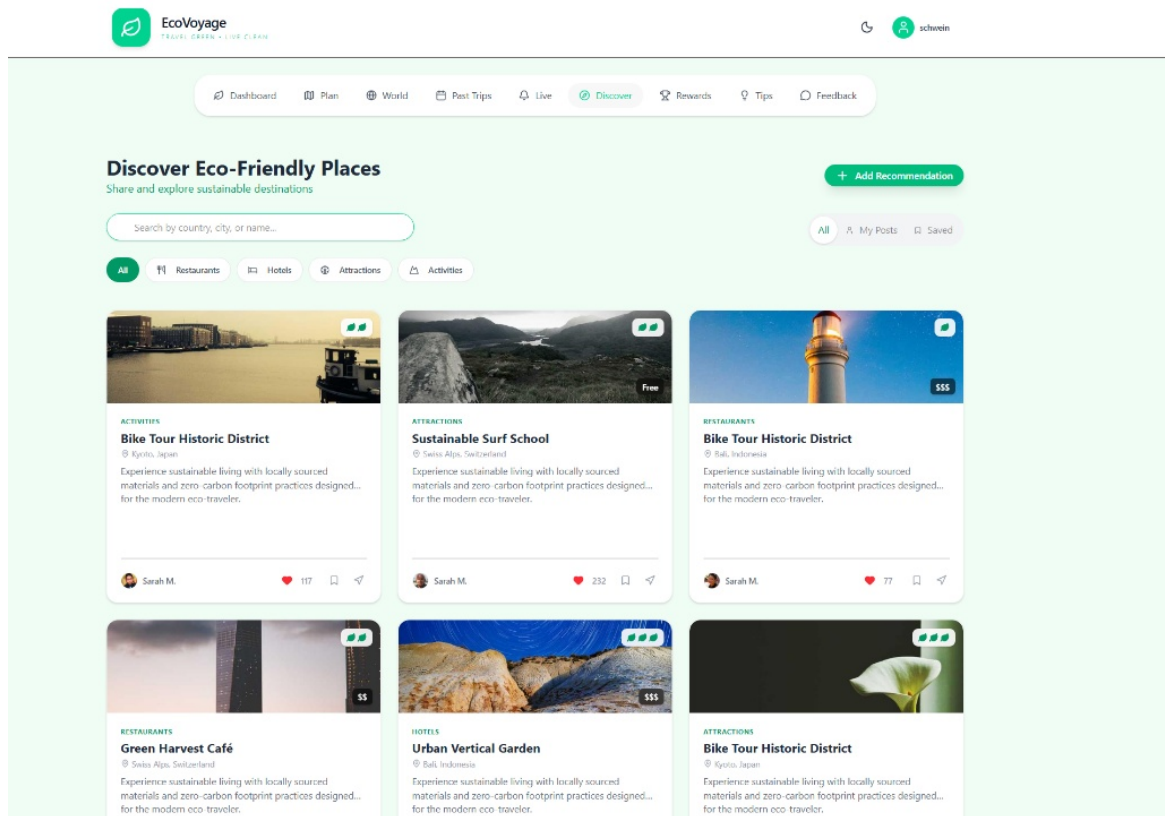


Figure 10: High-Fidelity Discover Interface



Eco Travel Planner

Plan Trip

Navigate City

+ Add Trip Segment

TRAVEL ☒ EAT/STAY

FROM

Search start location...

TO

Search destination...

MODE

Airplane

DATE

gg/mm/aaaa

FLIGHT DETAILS

Gate

--:--

--:--

DRAG MARKERS TO MAP



Add Segment

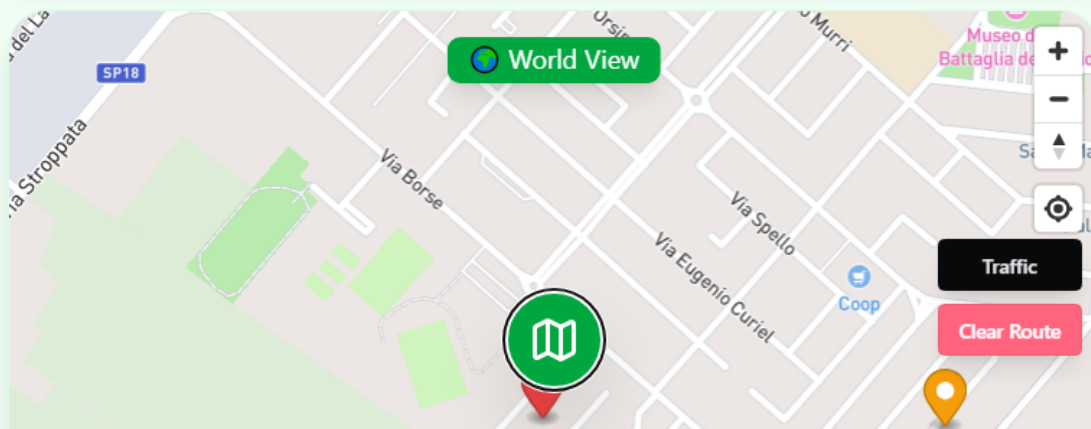


Figure 11: High-Fidelity Planning Interface

8 Usability Testing

8.1 Participants

We tested the app with 3 participants representing our user personas (students and frequent travelers).

8.2 Tasks Tested

1. **Discover:** Find a daily AI travel tip.
2. **Plan:** Plan a trip from Paris to Berlin and choose the lowest CO_2 option.
3. **Track:** Check Eco-Points and impact statistics.

Task	Success	Time	Key Issue
1. Discover (AI)	100%	30s	None.
2. Plan (Map)	100%	95s	Zooming on map felt sensitive.
3. Track (Stats)	100%	45s	None.

Table 1: Task Performance Metrics

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Task	Success	Time	Key Issue
1. Discover (AI)	100%	60s	None.
2. Plan (Map)	100%	50s	None.
3. Track (Stats)	100%	100s	Not very impressed by stats.

Table 2: Task Performance Metrics

Task	Success	Time	Key Issue
1. Discover (AI)	100%	10s	None.
2. Plan (Map)	100%	95s	Lagging.
3. Track (Stats)	50%	90s	First time didn't found dashboard.

Table 3: Task Performance Metrics

Figure 12: Tests results

8.3 Results Improvements

The tests were highly successful:

- **AI Tips:** Users loved the fresh content. We will add a loading indicator to manage the short wait time.
- **Planning:** Users appreciated the map but found zooming slightly sensitive.
- **Tracking:** Users liked the data but found the impact graph slightly hard to read, so we will simplify the visualization.

*For detailed results check **A5***

9 Conclusions

The EcoGo project successfully validated that users are willing to travel sustainably if provided with clear data and motivation. The **MEVN stack** proved robust for the application's needs. Our usability tests confirmed that **transparency** and **gamification** are key drivers for user engagement. Future work will focus on refining real-time notifications and smoothing out map interactions.

Detailed analysis and results are provided in A1, A2, A4, and A5.